STAFF REVIEW · H446 029 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q4

Do Now — source answers

1. ¬A A stays 0, so retain NOT A (¬A). B changes, so remove B.

2. [3, 6, 2, 8] Swap 8 and 3 → [3, 8, 6, 2]; swap 8 and 6 → [3, 6, 8, 2]; swap 8 and 2 → [3, 6, 2, 8]. The largest value is now last.

3. 24 values; GetLength(2) = 4 2 layers × 3 rows × 4 columns = 24 values. Dimension 2 is the third dimension (columns).

4. 01100000 0011 (denary 6) Align to exponent 3 by shifting the second mantissa right: 01000000 → 00100000. Add the mantissas: 01000000 + 00100000 = 01100000. The exponent remains 0011. This is normalised: $0.75 \times 2^3 = 6$.

I Do — source worked steps

1. Find the two pairs of 1s — ORANGE · TOP: circle columns 01 and 11 where A = 0. — BLUE · BOTTOM: circle columns 11 and 10 where A = 1. — Follow the same coloured groups on the map on the left. Each pair becomes one term.

2. Keep ¬A and C — A = 0 in both squares, so write ¬A . C = 1 in both, so keep C . B changes , so remove B.

3. Keep A and B — A = 1 in both squares, so keep A . B = 1 in both, so keep B . C changes , so remove C.

4. Join the two pairs with OR — Either coloured pair makes Q = 1 . AND the inputs inside each pair; OR the two results together.

5. Draw the actual logic gates — Trace it: A goes through NOT for the orange pair; A also goes directly to the blue pair. Each pair feeds an AND gate. OR the two AND outputs to get Q.

We Do — guided CHECK

1. 2. Compare your circles
The pair is on row A=1, across columns BC=01 and 11.
3. Simplify the group(s)
Use the two unchanged inputs to make one Boolean term. Which gate joins them?
4. Why is this correct?

2. 1. Find the 1s
Find the two largest pairs. Is it okay for the top-right 1 to be used twice?
2. Compare your circles
Top row and right column both contain a valid pair, sharing the top-right 1.
3. Simplify the group(s)
Top pair: A stays 0. Right pair: B stays 1. Write the two terms and join them.
4. Why is this correct?

3. 1. Find the 1s
The BC columns are 00, 01, 11, 10. Circle the four 1s at the two outside edges as ONE group.
2. Compare your circles
The outside columns (BC=00 and BC=10) are neighbours across the edge.
3. Simplify the group(s)
Across the four 1s, which variable stays 0? Which two change? Write Q.
4. Why is this correct?

4. 1. Find the 1s
Rows AB and columns CD use Gray code. Find two groups of four: top-left, then bottom-left and bottom-right across the edges.
2. Compare your circles
First group: top-left 2×2. Second group: the first and last columns in the bottom two rows form one wraparound group of four.
3. Simplify the group(s)
Write the fixed inputs for each group. Make each term, join with OR, and plan the gates.
4. Why is this correct?

You Do — source mark schemes

1. A=0 and B=0 throughout the pair; C changes.
Gate plan: Invert A and B, then AND them.

2. Top row: ¬A. Left column: ¬B. The top-left cell belongs to both groups.
Gate plan: Invert A and B, then OR them.

3. The two outside columns form ONE group of four. C=0; A and B change.
Gate plan: One NOT gate connected to C.

4. Top-right four: A=0 and C=1. Bottom outer four (across the edges): A=1 and D=0.
Gate plan: NOT A and NOT D; two AND gates feeding one OR gate.

Exit Ticket — source answer

1. True. Simplification must preserve the Boolean output for every input combination.